```
lis1=[10,20,30,40]
total=sum(lis1)
print("Sum of all items =  ",total)
```

```
Sum of all items =   100
```

Riya Kathe
1:05 PM Today

1. Write a Python program to sum all the items in a list.

```
num=[10,20,30,40]
smallest=num[0]
largest=num[0]

for n in num:
  if n<smallest:
    smallest=n
  elif n>largest:
    largest=n

print("Smsllest number = ",smallest)
print("largest number = ",largest)
```

```
Smsllest number =  10
largest number =  40
```

Riya Kathe
5:07 PM Today

2. Write a Python program to get the largest and smallest number from a list without builtin functions.

```
num = [10, 20, 30, 40, 20, 10, 50, 30]

duplicates = []
for n in num:
    if num.count(n) > 1 and n not in duplicates:
        duplicates.append(n)

print("Duplicate values are:", duplicates)
```

```
Duplicate values are: [10, 20, 30]
```

Riya Kathe
1:12 PM Today

3. Write a Python program to find duplicate values from a list and display those.

```
num = [1, 1, 2, 3, 4, 4, 5, 1]
n = 3

first_part = num[:n]
second_part = num[n:]

print("First part of the list:", first_part)
print("Second part of the list:", second_part)
```

```
First part of the list: [1, 1, 2]
Second part of the list: [3, 4, 4, 5, 1]
```

Riya Kathe
5:08 PM Today

4. Write a Python program to split a given list into two parts where the length of the first
part of the list is given.

```
list1=['red', 'green', 'white', 'black']
print(list(reversed(list1)))
```

```
['black', 'white', 'green', 'red']
```

Riya Kathe
5:11 PM Today

5. Write a Python program to traverse a given list in reverse order, and print the elements with the original index.

```
test_dict = {"A": 6, "B": 9, "C": 5, "D": 7, "E":

values = test_dict.values()
```

Riya Kathe
5:15 PM Today

```
mean = sum(values) / len(values)

print("Mean of dictionary values:", mean)
```

```
Mean of dictionary values: 6.2
```

1. Write a Python program and calculate the mean of the below dictionary.
test_dict = {"A" : 6, "B" : 9, "C" : 5, "D" : 7, "E" : 4}

```
dic1={1:10, 2:20}
dic2={3:30, 4:40}
dic3={5:50,6:60}

con1={**dic1,**dic2,**dic3}
print("3 dictionary Concatination = ",con1)
```

```
3 dictionary Concatination =  {1: 10, 2: 20, 3: 30
```

Riya Kathe
5:17 PM Today

2.Write a Python script to concatenate the following dictionaries to create a new one.
Sample Dictionary :
dic1={1:10, 2:20}
dic2={3:30, 4:40}
dic3={5:50,6:60}
Expected Result : {1: 10, 2: 20, 3: 30, 4: 40, 5: 50, 6: 60}

```
dict_num = {1: 10, 2: 20, 3: 30, 4: 40, 5: 50, 6:
print("key values")
for key,value in dict_num.items():
  print(key,'=',value)
```

```
key values
1 = 10
2 = 20
3 = 30
4 = 40
5 = 50
6 = 60
```

Riya Kathe
5:20 PM Today

3.Write a Python program to get the key, value and item in a dictionary.

```
input_dict = {1: 10, 2: 20, 3:None, 4: 40, 5: Non

print("Keys= ",input_dict.keys())
print("Values = ",input_dict.values())
print("Items = ",input_dict.items())
```

```
Keys=  dict_keys([1, 2, 3, 4, 5, 6])
Values =  dict_values([10, 20, None, 40, None, 60]
Items =  dict_items([(1, 10), (2, 20), (3, None),
```

Riya Kathe
5:23 PM Today

4.Write a Python program to get the key, value and item in a dictionary.

```
tuplex=(2, 4, 5, 6, 2, 3, 4, 4, 7 )
print("Number of times 4 appears = ",tuplex.count
```

```
Number of times 4 appears =  3
```

Riya Kathe
5:28 PM Today

1. Write a Python program to find the number of times 4 appears in the tuple.
Input:
tuplex = (2, 4, 5, 6, 2, 3, 4, 4, 7)
Output: 3

```
listx = [5, 10, 7, 4, 15, 3]
tup=tuple(listx)
print("Tuple = ",tup)
```

```
Tuple =  (5, 10, 7, 4, 15, 3)
```

Riya Kathe
5:31 PM Today

2.Write a Python program to convert a list to a tuple.
Input: listx = [5, 10, 7, 4, 15, 3]
Output: (5, 10, 7, 4, 15, 3)

```
tuples_list = [(1, 2), (3, 4), (5, 6)]

total = 0
for tup in tuples_list:
    for num in tup:
        total += num
```

Riya Kathe
5:32 PM Today

3. Write a Python program to calculate the sum of the numbers in a given tuple.
Input: tuples_list = [(1, 2), (3, 4), (5, 6)]

```
print("Sum of numbers in the tuple list:", total)
```

```
Sum of numbers in the tuple list: 21
```

Riya Kathe
5:35 PM Today

4.Write a python program and iterate the given tuples

```
employee1 =("John Deo",101,"Human Resource",60000
employee2 =("Alice Smith",102,"Marketing",55000)
employee3 =("Bob Johnson",103,"Engineering",70000

employees=[employee1,employee2,employee3]

for emp in employees:
  print("Name :",emp[0],"\n ID : ",emp[1],"\n Dep
```

```
Name : John Deo
 ID :  101
 Department :  Human Resource
 Salary :  60000
 ------------------------------
Name : Alice Smith
 ID :  102
 Department :  Marketing
 Salary :  55000
 ------------------------------
Name : Bob Johnson
 ID :  103
 Department :  Engineering
 Salary :  70000
 ------------------------------
```

Riya Kathe
5:45 PM Today

1. Write a Python program to Get Only unique items from two sets.

```
set1 = {10, 20, 30, 40, 50}
set2 = {30, 40, 50, 60, 70}

unique_items = set1.union(set2)

print("Unique items are:", unique_items)
```

```
Unique items are: {70, 40, 10, 50, 20, 60, 30}
```

Riya Kathe
5:52 PM Today

2. Write a Python program to Return a set of elements present in Set A or B, but not both.

```
set1 = {10, 20, 30, 40, 50}
set2 = {30, 40, 50, 60, 70}
set3=set1.symmetric_difference(set2)
print("Unique elements = ",set3)
```

```
Unique elements =  {20, 70, 10, 60}
```

Riya Kathe
5:59 PM Today

3. Write a Python program to Check if two sets have any elements in common. If yes, display the common elements.

```
set1 = {10, 20, 30, 40, 50}
set2 = {60, 70, 80, 90, 10}

common = set1.intersection(set2)

if common:
    print("Common elements are:", common)
else:
    print("No common elements found")
```

```
Common elements are: {10}
```

Riya Kathe
6:01 PM Today

```
set1 = {10, 20, 30, 40, 50}
set2 = {30, 40, 50, 60, 70}

set1 = set1.intersection(set2)

print("Updated set1:", set1)
```

```
Updated set1: {40, 50, 30}
```

3. Write a Python program to Remove items from set1 that are not common to both set1 and set2.